

Tom Naps Professional Biography

Thomas Naps is Professor of Computer Science at the University of Wisconsin Oshkosh, an undergraduate institution where he has been since 2001. Prior to that he was Associate Professor of Mathematics/Computer Science at Lawrence University from 1981 - 2001 and Assistant Professor of Mathematics at the Fox Valley campus in the University of Wisconsin College System from 1975 - 1981. He received his PhD in Mathematical Logic from the University of Notre Dame.

His professional development efforts during a 35-year teaching career have evolved from writing Computer Science textbooks, of which he co-authored 18 during the period 1982 - 2001, to the development of software for algorithm visualization (AV) and consequently incorporating that software into an active research program in Computer Science education (CSE). A key factor in this evolution from textbook author to AV and CSE researcher was a feature of his data structure textbooks called “graphic documentation” in which code was presented side-by-side with detailed depictions of how that code affected the data structures. This desire to explain the actions of algorithms by graphically portraying them led to his designing the GAIGS and JHAVÉ AV systems (jhave.org). JHAVÉ is a pedagogical environment for delivering AVs that provides a collection of tools designed to ensure students do not just passively watch the visualization but instead actively engage with it. These features include: context-sensitive HTML documentation windows that provide textual explanations of the depicted snapshots in a visualization, pseudo-code windows that link the visualization of the algorithm to synchronized pseudo-code for the algorithm, input generators that encourage “what if” exploration by allowing students to provide different data sets to the algorithm, stop-and-think questions that pause the visualization and ask the student a question that makes them predict the next step of the algorithm’s execution, and integration with a database that allows registered students to have their responses to stop-and-think questions logged for instructor evaluation.

The jhave.org site provides short tutorials on a variety of algorithms depicted by JHAVÉ, including seven sorting algorithms, eight graph algorithms, four hashing algorithms, and eight other miscellaneous algorithms. Since 2007, JHAVÉ has delivered on average over 2000 instructional visualizations per year to learners in over 50 countries and has been used by instructors at over 60 different universities.

More recently Dr. Naps has directed his efforts at working with a growing group of content authors and software developers on the OpenDSA project, which is attempting to build an online open-source interactive eTextbook on data structures and algorithms.

The merging of his work in these software development projects with efforts to demonstrate how such software can be effectively used in instruction has led to his being author or co-author of three journal articles and 26 conference articles. Additionally it has resulted in eleven major grants (ten from the National Science Foundation and one from Ameritech), eight professional workshops presented on AV, and the mentoring of numerous undergraduates in research projects, fifteen of which have resulted in their presenting at professional conferences.

His ability to collaborate effectively with other researchers in the CSE community has led to his co-chairing seven ITiCSE working groups and participating in two others. The report of the group he co-chaired Guido Rößling in 2002 – “Exploring the role of visualization and engagement in computer science education” – has proved to be influential in establishing a taxonomy of the ways in which learners can actively engage with algorithm visualizations. It has been cited over 400 times, giving strong indication that it has helped others in the CSE community in designing a variety of research efforts that explore how visualization can lead to better learning outcomes for students.

Beyond his collaborations in such research efforts, Dr. Naps has been active in serving the SIGCSE community, having been program co-chair for the SIGCSE symposium in 2004, program co-chair for the ITiCSE conference in 2011, and SIGCSE symposium co-chair in 2005. In 2013, he was one of four ACM members who received the distinction of Distinguished Educator.